



ENDOCRINOLOGY

Pituitary Gland - 2

OBJECTIVES

- Understand what is prolactinoma, acromegaly & diabetes insipidus.
- Know how to diagnose & treat these diseases.

The AIM of this lecture is:

1. Understand what is prolactinoma, acromegaly & diabetes insipidus.
2. Know how to diagnose & treat these diseases.

By the end of this lecture the student will be able to:

1. Define prolactinoma, acromegaly & diabetes insipidus.
2. Be able to do a proper approach to each condition.

Hyperprolactinaemia/galactorrhoea

- Hyperprolactinaemia is a common abnormality that usually presents with **hypogonadism** and/or **galactorrhoea** (lactation in the absence of breastfeeding).
- Since prolactin stimulates milk secretion but not breast development, galactorrhoea rarely occurs in men and only does so if gynaecomastia has been induced by hypogonadism.
- Many drugs, especially dopamine antagonists, elevate prolactin concentrations.
- Pituitary tumours can cause hyperprolactinaemia by directly secreting prolactin (prolactinomas), or by compressing the infundibular stalk and thus interrupting the tonic inhibitory effect of hypothalamic dopamine on prolactin secretion ('disconnection' hyperprolactinaemia).

20.57 Causes of hyperprolactinaemia	
Physiological	
- Stress (e.g. post-seizure) - Pregnancy - Lactation - Nipple stimulation	- Sleep - Coitus - Exercise - Baby crying
Drug-induced	
Dopamine antagonists	
- Antipsychotics (phenothiazines and butyrophenones) - Antidepressants	Antiemetics (e.g. metoclopramide, domperidone)
Dopamine-depleting drugs	
Reserpine	Methyl dopa
Oestrogens	
Oral contraceptive pill	
Pathological	
Common	
- Disconnection hyperprolactinaemia (e.g. non-functioning pituitary macroadenoma) - Prolactinoma (usually microadenoma)	- Primary hypothyroidism - Polycystic ovary syndrome
Uncommon	
Pituitary tumour secreting prolactin and growth hormone	- Hypothalamic disease - Renal failure
Rare	
Chest wall reflex (e.g. post herpes zoster)	

In women , in addition to galactorrhoea, hypogonadism associated with hyperprolactinaemia causes secondary amenorrhoea and anovulation with infertility.

Important points in the history include drug use, recent pregnancy and menstrual history.

In men there is decreased libido, reduced shaving frequency and lethargy.

Unilateral galactorrhoea may be confused with nipple discharge, and breast examination to exclude malignancy or fibrocystic disease is important.

Investigations:

- **Pregnancy should first be excluded** before further investigations are performed in women of child-bearing potential.
- Patients with prolactin excess should have tests of gonadal function and T4 and TSH should be measured to exclude primary hypothyroidism causing TRH-induced prolactin excess.
- Unless the prolactin falls after withdrawal of relevant drug therapy, a serum prolactin consistently above the reference range is an indication for MRI or CT scan of the hypothalamus and pituitary.
- Patients with a macroadenoma also need tests for hypopituitarism.

Management:

- If possible, the underlying cause should be corrected (e.g. cessation of offending drugs and giving levothyroxine replacement in primary hypothyroidism).
- If dopamine antagonists are the cause, then dopamine agonist therapy is contraindicated; if gonadal dysfunction is the primary concern, sex steroid replacement therapy may be indicated.
- Troublesome physiological galactorrhoea can also be treated with **dopamine agonists**

Most prolactinomas in pre-menopausal women are microadenomas because the symptoms of prolactin excess usually result in early presentation.

Prolactin-secreting cells of the anterior pituitary share a common lineage with GH-secreting cells, so occasionally prolactinomas can secrete excess GH and cause acromegaly.

In prolactinomas there is a relationship between prolactin concentration and tumour size: [the higher the level, the bigger the tumour](#).

Some macroprolactinomas can elevate prolactin concentrations above 100 000 mIU/L (4700 ng/mL). The investigation of prolactinomas is the same as for other pituitary tumours

Management:

1. (medical) **Dopamine agonist drugs are first-line therapy** for the majority of patients, They usually reduce serum prolactin concentrations and cause significant tumour shrinkage after several months of therapy.

20.58 Dopamine agonist therapy: drugs used to treat prolactinomas			
Drug	Oral dose*	Advantages	Disadvantages
Bromocriptine	2.5–15 mg/day 2–3 times daily	Available for parenteral use Short half-life; useful in treating infertility Proven long-term efficacy	Ergotamine-like side-effects (nausea, headache, postural hypotension, constipation) Frequent dosing so poor adherence Rare reports of fibrotic reactions in various tissues
Cabergoline	250–1000 µg/week 2 doses/week	Long-acting, so missed doses less important Reported to have fewer ergotamine-like side-effects	Limited data on safety in pregnancy Associated with cardiac valvular fibrosis in Parkinson's disease
Quinagolide	50–150 µg/day Once daily	A non-ergot with few side-effects in patients intolerant of the above	Limited data on safety in pregnancy

*Tolerance develops for the side-effects. All of these agents, especially bromocriptine introduced at low dose and increased slowly. If several doses of bromocriptine are introduced, the process must start again.

a macroprolactinoma has failed to shrink sufficiently with dopamine agonist therapy, and this may be because the tumor has a significant cystic component.

Surgery may also be performed in patients who are intolerant of dopamine agonists.

External irradiation may be required for some macroadenomas to prevent regrowth if dopamine agonists are stopped.

Special situation !

Pregnancy

Hyperprolactinaemia often presents with infertility, so dopamine agonist therapy may be followed by pregnancy.

Patients with microadenomas should be advised to withdraw dopamine agonist therapy as soon as pregnancy is confirmed.

In contrast, macroprolactinomas may enlarge rapidly under oestrogen stimulation and these patients should continue dopamine agonist therapy and need measurement of prolactin levels and visual fields during pregnancy.

All patients should be advised to report headache or visual disturbance promptly.

Acromegaly

Acromegaly is caused by growth hormone (GH) secretion from a pituitary tumour, usually a macroadenoma, and carries an approximate two fold excess mortality when untreated.

Clinical features

If GH hypersecretion occurs before puberty, then the presentation is with gigantism.

More commonly, GH excess occurs in adult life and presents with acromegaly.

If hypersecretion starts in adolescence and persists into adult life, then the two conditions may be combined. The most common complaints are headache and sweating. Additional features include those of any pituitary tumour. **Rarely Acromegaly may be caused by**

ستاجير

كل ما يحتاجه في الطب

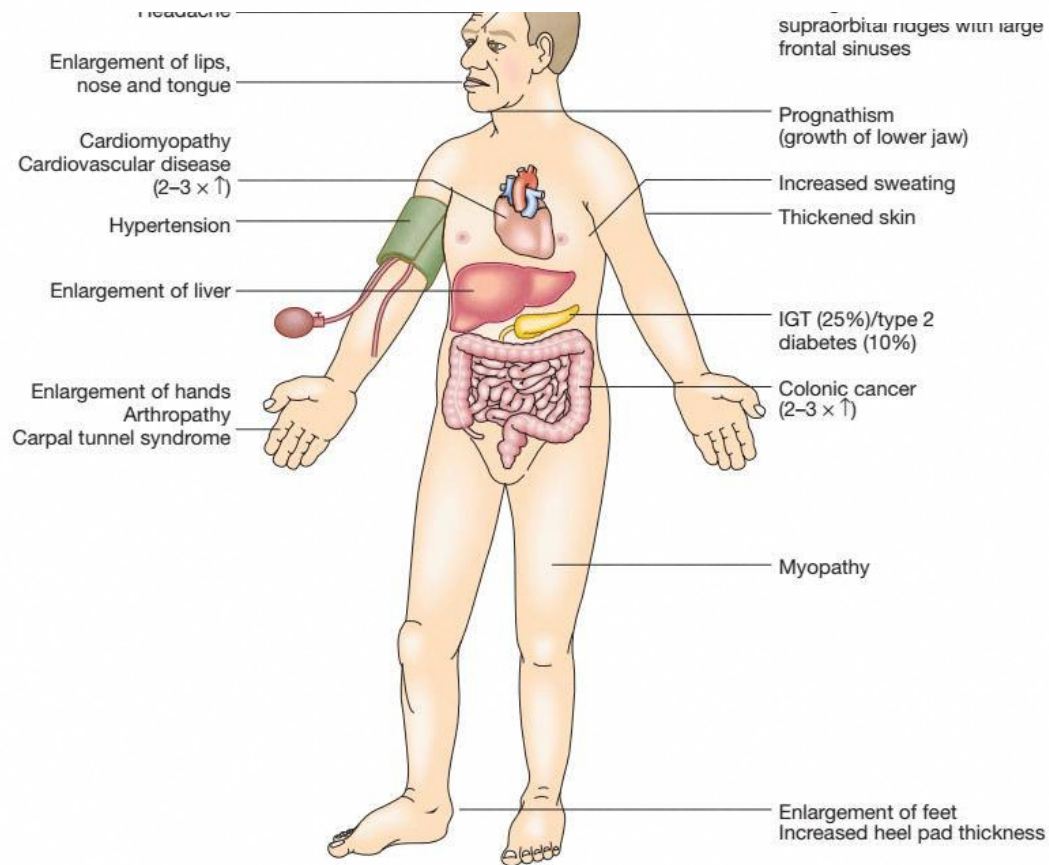
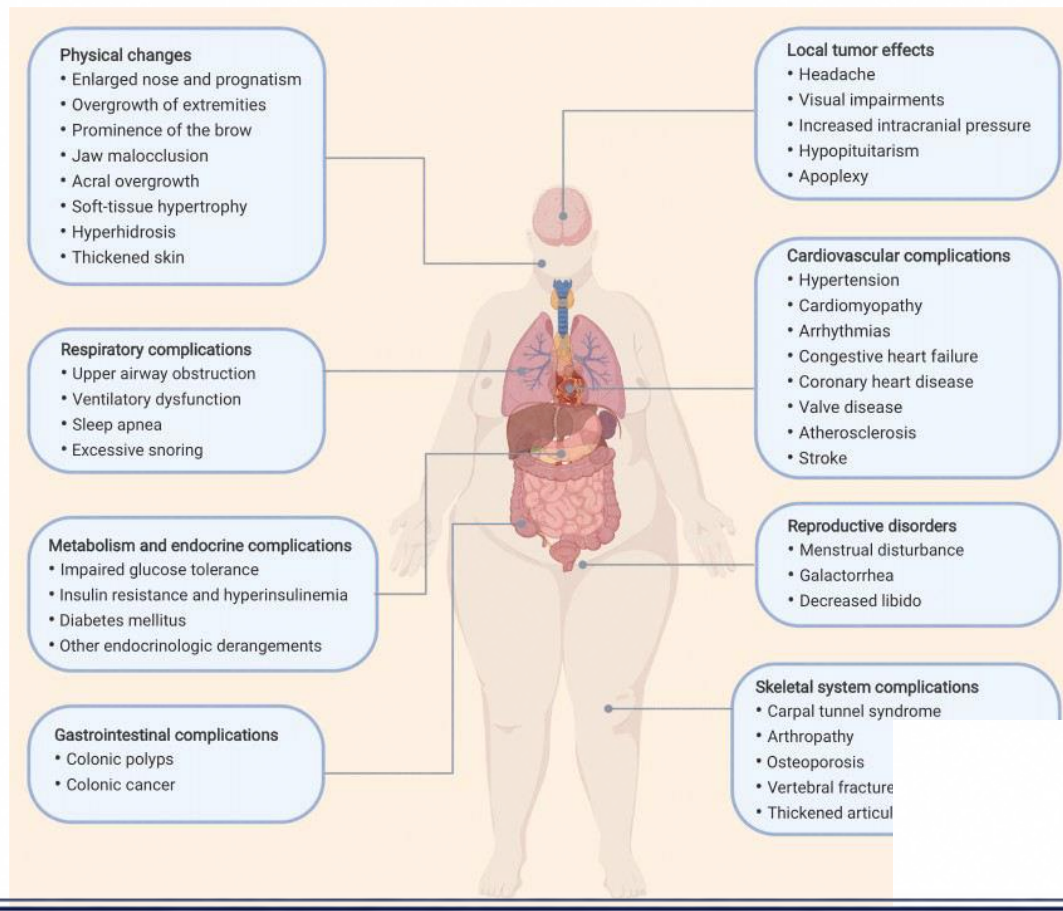


Fig. 20.29 Clinical features of acromegaly. (IGT = impaired glucose tolerance)





- The clinical diagnosis must be confirmed by measuring GH levels during an oral glucose tolerance test and measuring serum IGF-1.
- In normal subjects, plasma GH suppresses to below 0.5 µg/L (approximately 2 mU/L).
- In acromegaly, GH does not suppress and in about 30% of patients there is a paradoxical rise; IGF-1 is also elevated.
- The rest of pituitary function should be investigated.
- Prolactin concentrations are elevated in about 30% of patients due to co-secretion of prolactin from the tumour.
- Additional tests in acromegaly may include screening for colonic neoplasms with colonoscopy.

Management

1. **Surgical:** Trans-sphenoidal surgery is usually the first line of treatment and may result in cure of GH excess, especially in patients with microadenomas. More often, surgery serves to debulk the tumour and further second-line therapy is required, according to post-operative imaging and glucose tolerance test results.
2. **Radiotherapy:** External radiotherapy is usually employed as second or third line treatment if acromegaly persists after surgery, to stop tumor growth and lower GH levels. However, GH levels fall slowly (over many years) and there is a risk of hypopituitarism
3. **Medical:** If acromegaly persists after surgery, medical therapy is usually employed to lower GH levels to below 1.5 µg/L (below approximately 5 mU/L) and to normalize IGF-1 concentrations.

Medical therapy may be discontinued after several years in patients who have received radiotherapy.

Somatostatin analogues (such as octreotide, lanreotide or pasireotide) can be administered as slow-release injections every few weeks.

alternative or in advance of surgery, given evidence that they can induce modest tumor shrinkage in some patients.

- **Dopamine agonists** are less effective at lowering GH but may sometimes be helpful, especially with associated prolactin excess.
- **Pegvisomant** is a peptide GH receptor antagonist administered by daily self-injection and may be indicated in some patients whose GH and IGF-1 concentrations fail to suppress sufficiently following somatostatin analogue therapy.

Diabetes insipidus

characterised by the persistent excretion of excessive quantities of dilute urine and by thirst. It is classified into two types:

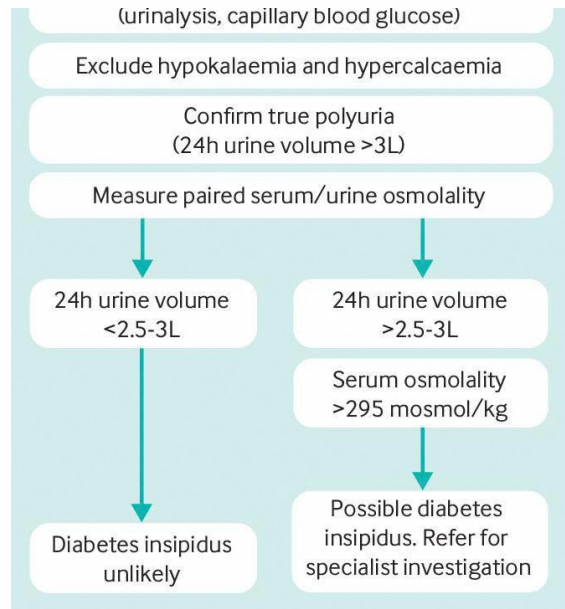
1. **cranial diabetes insipidus**, in which there is deficient production of vasopressin by the hypothalamus
2. **Nephrogenic diabetes insipidus**, in which the renal tubules are unresponsive to vasopressin.

i	20.59 Causes of diabetes insipidus
Cranial	
Structural hypothalamic or high stalk lesion	
• See Box 20.53	
Idiopathic	
Genetic defect	
• Dominant (<i>AVP</i> gene mutation) • Recessive (DIDMOAD syndrome – association of diabetes insipidus with diabetes mellitus, optic atrophy, deafness)	
Nephrogenic	
Genetic defect	
• V2 receptor mutation • Aquaporin-2 mutation • Cystinosis	
Metabolic abnormality	
• Hypokalaemia • Hypercalcaemia	
Drug therapy	
• Lithium • Demeclocycline	
Poisoning	
• Heavy metals	
Chronic kidney disease	
• Polycystic kidney disease • Sickle-cell anaemia • Infiltrative disease	

- The most marked symptoms are **polyuria and polydipsia**.
- The patient may pass 5–20 L or more of urine in 24 hours.
- This is of low specific gravity and osmolality.
- If the patient has an intact thirst mechanism, is conscious and has access to oral fluids, then he or she can maintain adequate fluid intake.
- However, in an unconscious patient or a patient with damage to the hypothalamic thirst Centre, diabetes insipidus is potentially lethal.
- If there is associated cortisol deficiency, then diabetes insipidus may not be manifest until glucocorticoid replacement therapy is given.
- The most common differential diagnosis is primary polydipsia, caused by drinking excessive amounts of fluid in the absence of a defect in ADH or thirst control.

Investigations

- Diabetes insipidus can be confirmed if **serum vasopressin is undetectable (although the assay for this is not widely available) or the urine is not maximally concentrated (i.e. <600 mOsmol/kg) in the presence of increased plasma osmolality (i.e. >300 mOsmol/kg).**
- Sometimes, the diagnosis can be confirmed or refuted by random simultaneous samples of blood and urine, but more often a dynamic test is required.
- The water deprivation test is widely used :
- (Diabetes insipidus is confirmed by a plasma osmolality >300 mOsmol/kg with a urine osmolality <600 mOsmol/kg
- » **Cranial diabetes insipidus** is confirmed if urine osmolality rises by at least 50% after DDAVP
- » **Nephrogenic diabetes insipidus** is confirmed if DDAVP does not concentrate the urine
- » **Primary polydipsia** is suggested by low plasma osmolality at the start of the test



Management:

- Treatment of cranial diabetes insipidus is with desamino-des-aspartate-arginine **vasopressin (desmopressin, DDAVP)**, an analogue of ADH which has a longer half-life.
- DDAVP is usually administered **intranassally**.
- An oral formulation is also available but bioavailability is low and rather unpredictable.
- In sick patients, DDAVP should be given by intramuscular injection.
- The dose of DDAVP should be adjusted on the basis of serum sodium concentrations and/or osmolality.
- The principal hazard is excessive treatment, resulting in water intoxication and hyponatraemia.
- Conversely, inadequate treatment results in thirst and polyuria.
- The ideal dose prevents nocturia but allows a degree of polyuria from time to time before the next dose.
- **The polyuria in nephrogenic diabetes insipidus is improved by thiazide diuretics** (e.g. bendroflumethiazide 5–10 mg/day), amiloride (5–10 mg/day) and NSAIDs (e.g. indometacin 15 mg 3 times daily), although the last of these carries a risk of reducing glomerular filtration rate